EMUNOT VEDEOT: SAADIAH GAON'S BOOK OF BELIEFS AND OPINIONS

Can Reason Prove the World Had a Beginning?

Emunot VeDeot, Essay 1:1

Sefer: Emunot VeDeot (The Book of Beliefs and Opinions) by Rav Saadia Gaon **Essay/Chapters:** Essay 1 (Creation), Chapter 1, Part 1: the scriptural sources and the first three of Rav Saadia's four proofs **Shiur:** Part of Dayan Daniel Kada's Monday series on Emunot VeDeot, Monday 8 April 2024, on Zoom

These notes are based on an automatic transcript of the recording.

Where We Are

The class had finished **the whole Introduction** (a summary is on the WhatsApp group) and begun the introduction to **Essay 1 of 10**, on **how creation came about**. The essay has four chapters:

Chapter	Content
1	**Four proofs** for the creation of the universe (tonight: the first three)
2	Develops this further
3-4	**Twelve other opinions**, examined one by one, and related questions

Why this is hard to prove

In his introduction to the essay Rav Saadia admitted the difficulty. Of the three sources of knowledge from the book's Introduction:

1. **Sense perception:** nobody was around to see creation.
2. **Self-evident axioms:** creation is not one.
3. **Deduction:** it is very hard to deduce something you have no sense perception of.

His answer: **don't worry**, because every rival theory has the same problem and must also rely on deduction. "Actually, you should know, my proofs are going to be stronger, so trust what I'm going to say."

"Very courageous of Saadia, to put forward his view with his proofs, then list all the other opinions, twelve of them, disprove them, and say: you know, mine are the best. Well, let's go through it and see if he's right."

Part 1: Why the Rabbi Will Move Quickly Through Essay 1

"In essence, this whole essay we could skip if we wanted to, because **contemporary science has now proven that the world did have a beginning**."

Rav Saadia believes in **creation *ex nihilo***, creation from nothingness: there was a "time" when nothing existed except the first existence, God, called in medieval literature **the First Existent (*HaMatzui HaRishon*)**, the First Cause. **How long ago** is not the question here: 5,700 years or billions of years "makes no difference." The claim is only that **there was a beginning**.

The three classical positions

View	Claim	Who held it
Aristotle	God and the universe are **both eternal**; they always existed and always will	Aristotle
Plato	God created the world, but **from eternal matter (*hyle*, היולי)** that always coexisted with Him	Plato; among the Rishonim, **Rabbi Avraham ibn Ezra**
Creation from nothing	There was nothing but God, and He created everything	**Rav Saadia** and the majority of the Rishonim

The Rambam's struggle with Aristotle

The Rambam loved Aristotle, "one of his main philosophical teachers," and was troubled that Aristotle held the world to be eternal. In the **Moreh Nevuchim**:

- The Rambam brings **seven proofs** for creation, **three of which are among Rav Saadia's four**.
- His famous conclusion: **Aristotle's arguments are not conclusive, so I have no need to accept them.**
- **If Aristotle had proven the eternity of the world, I would have reinterpreted the Torah to fit it.** And that, he says, would have been **easier** than reinterpreting all the verses that seem to give God a body.

On **Plato**, the Rambam wrote that he himself **prefers creation from nothing**, but that **those who accept Plato's view have something to rely on**: it can be made compatible with the Torah. The ibn Ezra read **בראשית ברא אלהים** that way: God created from eternal "floating matter," and that matter is not itself called a creation.

Rav Saadia rejects them all. "Saadia is going to say that it is our belief that creation is from nothingness."

Science and *Yehi Or*

Today the debate is settled: **the universe most definitely had a beginning.**

- **Redshift**: the universe is continuously expanding, which means there was something at the beginning, however small, from which it all began.
- This fits **the Big Bang**, "which is completely compatible with Torah."
- **The Ramban** at the beginning of Bereshit seems to indicate something like it. Chazal say **everything was created on day one** and only laid out over the six days. **ויאמר אלהים יהי אור**: the famous question of how there could be light before the sun on day four. One basic answer: ***Yehi Or* was the light of the Big Bang**, the initial spark from which everything shot out.

"So I'm going to go slightly fast through essay one, because we know the truth, both from science and Torah." But it is still worth learning "how the medieval Hachamim used their logic and their philosophy."

Part 2: The Thesis and the Verses

Hashem has told us that **everything is created, and that He created it from nothing**, as it says **בראשית ברא אלהים את השמים ואת הארץ**.

The word *bara*

"A famous argument brought by the Rishonim, and Saadia may have been the first to put it forward":

The word ברא is used only of God. Nowhere in Tanach is *bara* used of a human being. For people we find **יצר (*yatzar*, formed)** and **בנה (*banah*, built)**.

Word	Meaning
יצירה (*yetzirah*)	**Forming** from what already exists. A **צייר (*tzayar*)**, an artist, takes what is in front of him and forms it
בריאה (*beri'ah*)	A more intense kind of creation: **creation from nothing**

The Rabbi connected this to the **four worlds of Kabbalah**: **Asiyah** (action, this world), then **Yetzirah**, then **Beri'ah** higher still. Beri'ah is higher than Yetzirah "because *yetzirah* is forming from existence; *beri'ah* is creating from nothing."

Rav Saadia also brings Yeshayahu:

> I am Hashem who makes all things, who alone (Yeshayahu 44:24) **אנכי ה' עשה כל נטה שמים לבדי** stretched out the heavens.

Then philosophy

Since the Torah has told us this and it is verified by signs and wonders, **we accept it**. *Then* Rav Saadia asks whether it can be **proven rationally**.

"This is the structure we said he would follow in each essay: put forward the theory, prove it from the verses, and then try to do it philosophically."

If philosophy can prove it, why do we need the Torah? He answered this in the Introduction: **some people cannot do it philosophically**, so the Torah gives the facts. **Our duty**, once we have accepted what the Torah says, is to analyse it and see whether we can prove it philosophically too.

"There are many proofs. I'm only going to give you four."

Part 3: Proof One: The Finite Universe

Heaven and earth are **finite** in size. **A finite body can only contain a finite force.** And since the force within them is finite, **they must have a beginning and an end.**

The argument as the Rabbi laid it out:

1. **The earth is in the middle and the heavens revolve around it.** (Rav Saadia's geocentric astronomy.)
2. So the whole system is **finite**.
3. Everything has a **force** within it that sustains it; if the body is finite, its force must be finite.
4. A finite force eventually stops, **and what has an end must have had a beginning.**

Rav Saadia tests his own proof

"When this proof was told to me, **I didn't speedily accept it until I had verified it.**"

He raises three objections against himself:

Objection	Rav Saadia's reply
Perhaps the earth is infinite in length and breadth	Then **the sun could not go around it every day** and return to where it began, nor could the moon and stars
Perhaps the earth is finite but the heavens infinite	We **see** the heavens rotating around us. Something that goes round has a circumference, so it is finite
Perhaps what rotates is finite, but beyond it there is more, static and endless	**Why assume something we have no perception of?** We see nothing beyond what rotates around us

Is the proof still any good?

The obvious problem: **it is based on a geocentric universe**, which we now know is wrong. But the Rabbi argued that a version survives in a heliocentric universe: "Just do it the other way round. How could the earth go around the sun if it were infinite?" A rotation means a circumference, and a circumference means it is finite. A participant, Mark, agreed: **whichever way you look at it, a rotation means an end.** (When asked, the Rabbi clarified that this proof is about **space**; **time** is the fourth proof.)

He also cautioned against reading Rav Saadia **anachronistically**: "Now we know there's much more beyond what we see than the millions of galaxies. But in those days all you had is the earth and the things around it. Why should you assume there's more behind it?" For them, **the only universe was the observable universe**.

"Look, don't forget, we're talking about someone in the 900s. The science is going to be flawed, obviously. And he would be the first one, I'm sure, like the Rambam would be, to say: accept the science of the day. But the arguments can still somewhat stand."

Many worlds?

Then Rav Saadia raises something remarkable: **perhaps there are infinitely many earths and heavens**, each with its own observable rotation, going on forever in space and time. "It's interesting that he thought of this."

His rejection is **scientific**, not theological:

By nature **earth cannot be above fire**, and earth will not sit above air or water, because fire and air are lighter, and earth and water heavier. If there were another earth somewhere else, **it would**

fall through the air and fire and land on our earth; another mass of water would come down through the atmosphere onto ours.

"They didn't know about gravity," someone noted. Exactly, said the Rabbi: "It's very interesting. He was willing to accept there may be other universes, but felt it would be scientifically incorrect. Which means that nowadays, when this scientific reason is no longer true, then perhaps." He added that a later **Spanish Jewish philosopher from Zaragoza** concluded that **parallel universes are possible**, "why on earth should we be so arrogant as to assume that all we have is our universe?"

The verses

From one end of the heavens to the other end of (Devarim 4:32) **ולמקצה השמים ועד קצה השמים** .the heavens

Moshe is exhorting the people: has any nation, from one end of the earth to the other, witnessed anything like Har Sinai? "Clearly that's just exhorting the people," but Rav Saadia reads in it that **heaven and earth have an end**. "It's an interesting proof. I don't know how convincing it is."

.The sun rises and the sun sets, and returns to its place (Kohelet 1:5) **וזרח השמש ובא השמש**

Rav Saadia takes this as testimony that **the sun circles the earth daily**.

Galileo and "sunrise"

The Rabbi noted that **this verse was the source of the Church's opposition to Copernicus and Galileo**: the verse says the sun rises and sets, so the sun moves. Does that make our Torah wrong?

"Let me ask you a question. Now that we have Copernicus and Galileo, **why do we still talk about sunrise and sunset?** There's no such thing! The answer is: that's what we see." The verse speaks the same way, **in the language of what we see.**

"Saadia took this further and actually believed from it that the sun rotates around the earth. He was wrong. That was the science at the time." But the crux of the argument, he said, can still be accepted in some version.

Part 4: Proof Two: Composition and Design

"This is essentially what we call nowadays in philosophy **the argument from design**, or the teleological argument," though Rav Saadia does not express it in those words.

The second proof: from the joining of parts and the putting together of things. I look at the whole world and see that **everything is beautifully put together**, and it is clear to me that **there must be a Creator**.

Again he tests it step by step:

1. **Perhaps composition is only in living bodies**, mammals and plants, but not in larger things? No: look at **the earth**: **soil, stone and sand**, all combined together.
2. **Perhaps not in the heavens?** "Then I go up to heaven in my mind" and see **sphere within sphere**, all within each other, some stars brighter and some dimmer, forming one magnificent cosmos. (The astronomy of the time pictured the planets nested **like onion shells**, each inside the other. "Obviously we know that's not true nowadays.")

"When I saw all of this composition, I accepted the second proof."

The verses:

ידיך עשוני ויכוננוני (Tehillim 119:73): Your hands made me and established me. The way my body is put together shows an artisan.

יוצר הארץ ועשה הוא כוננה (Yeshayahu 45:18): He formed the earth and made it, He established it.

כי אראה שמיך מעשה אצבעתיך ירח וכוכבים אשר כוננתה (Tehillim 8:4): when I see Your heavens, the work of Your fingers, the moon and the stars which You established.

What it is meant to defeat: the claim that the world is eternal. "If it went on and on and on, fine. But how did this amazing composition come together?"

The quill and the dog

This became **"the most famous and strongest proof of the medieval Rishonim"** after Rav Saadia. The Rambam used it, and it was developed by a philosopher from Zaragoza in Spain with an example used by many after him:

You leave a **quill, ink and a blank sheet of parchment**. You come back a day later and find **a beautiful piece of writing**. Either someone wrote it, or **a dog knocked the ink over** and out came beautiful script. "That is possible, 0.001 percent, but it's highly unlikely. **The design shows.**"

Rav Shach and the eye surgeon. Rav Shach in Israel had a major eye operation. Afterwards the surgeon showed him everything they had done and how intricate the eye is. Rav Shach turned to him and said: "How can you not believe, after doing what you did? How can you not believe there's a Creator, that someone put all of this together?"

Julian and the second law of thermodynamics

A participant, Julian, connected the proof to modern physics: **the second law of thermodynamics (entropy)** says order in the world is always decreasing. "If I show you a film of an **ice cream melting** and then show it backwards, you know which way is forwards, because an ice cream doesn't freeze itself from drips back into a cone." Creating more order requires work, energy put in. "We have this massive system of order and detail which can't happen by itself."

The Rabbi's caveats:

- Rav Saadia **assumes God**. He is not proving God here (in his time everyone accepted a Creator; the dispute was about the right way to serve Him). He is proving **creation**, a beginning.
- The design argument **does not disprove Plato**: "You could still say, yes, Hashem designed it, but there was still eternal matter."

"That, I think, is a very strong proof."

Part 5: Proof Three: Accidents (*Mikrim*)

"This is **Kalam** philosophy now," the method of the Arabic theologians (the Ash'arites among them).

Not "accidents" in the modern sense

In the translations the term is **"accidents"** (Hebrew **מקרים, *mikrim***). "Be careful you don't get confused. It's not accident in the sense of what we call an accident today." It means **happenings**: things that happen to a body. **Every happening is a new thing.**

Nothing physical stays static. Even a table: plastic may take hundreds or thousands of years, "but it eventually decays." Things grow, are completed, and break down; things happen to them from within and from without.

The argument

1. **Every physical body has accidents happening to it** constantly, internally or externally.
2. **Every accident is a new, created thing.**
3. **A body is never free of its accidents.**
4. **Therefore the body itself is also created.**

"Since the happenings to these bodies are all new things, **the thing itself must also be new**, must also have had a creation."

The Rabbi admitted: "I'll be honest, I did find this difficult to get my head around. I initially thought the argument was that since everything decays, it must have had a beginning. **He's not saying that.**" The point is that whatever is inseparable from created things is itself created. Julian called it a complete argument, "very complex, and it takes a while to get your head around it."

A participant suggested **a seed**: it grows and dies. The Rabbi pushed back that one could still ask: "Maybe the seed was there for eternity, and things just happen to it all the time: it gets watered and grows and decays, and grows and decays." That is exactly what the proof must exclude.

Testing it on the earth and the heavens

Objection	Reply
Perhaps the earth itself has no accidents, only the plants on it?	The earth is **inherently connected** with what grows and changes on it, so it follows the same laws: it too is a happening
Perhaps the heavens have no accidents?	They have them constantly: **perpetual motion** that never rests; some move **fast, some slow**; some **give light** to others and some **receive light** (like the moon); **different colours**

So the heavens, too, are created. The verse:

> **אנכי עשיתי ארץ ואדם עליה בראתי, אני ידי נטו שמים** (Yeshayahu 45:12) I made the earth and created man upon it; My hands stretched out the heavens.

A striking reversal: Aristotle held that the perpetual rotation of the heavenly bodies **proves they are eternal**. Rav Saadia reads the very same motion the other way: because things are always happening to the heavens, **they are happenings themselves, and had a beginning**.

Questions from the Class

"When was this written, and how did they know what was going on in space? They didn't have telescopes." (Maria)

About a thousand years ago. "They had a whole science at the time," including the 28 stations of the moon. Rav Saadia was not a scientist writing in the Torah; he was "the big rabbi, the first," writing on the basis of the science of his day. "Of course the science is going to be flawed. That's not the point. We're looking at **the way he's thinking**, and some of the arguments are still applicable." Why God created all the stars: "a question for another time."

"Don't diamonds last forever?"

"That's why the song goes like that." But **all physical matter decays**; some faster, some slower. "It may take thousands and thousands of years, but everything physical comes to an end." Someone added: even uranium.

Summary

1. **Essay 1** argues for **creation from nothing**. Since science now confirms the universe had a beginning (redshift, the Big Bang, perhaps hinted at by the **Ramban** on *Yehi Or*), the Rabbi will move through it relatively quickly.
2. **Three positions: Aristotle** (eternal universe), **Plato** (creation from eternal matter, accepted as legitimate by the Rambam and held by the **ibn Ezra**), and **creation from nothing** (Rav Saadia and most Rishonim). The **Rambam**: had Aristotle proven eternity, he would have reinterpreted the Torah.
3. **Scripture first: *bara*** is used only of God and means creation from nothing, as distinct from ***yatzar***.
4. **Proof 1, finitude:** heaven and earth are finite, so their force is finite, so they began. Built on geocentric astronomy, but the Rabbi thinks a version survives. Rav Saadia even considers **many worlds**, rejecting them on the physics of his day.
5. **Proof 2, design:** everything is composed and put together, so there is a Maker (the **quill and the dog**, **Rav Shach**, and **entropy**). It does not, on its own, refute Plato.
6. **Proof 3, accidents:** whatever is never free of created happenings is itself created, applied to the earth and even to the heavens, against Aristotle.

Coming Up

Next week (6 o'clock): the **fourth proof, from time**.

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